

- the formation of gametes as a result of meiosis
- the process of fertilisation.

- Explain why the cell that is formed as a result of fertilisation is a diploid cell **and** contains homologous chromosomes.

[3]

- stage in meiosis
- reason
-
-

[2]

- (c) Microscope slides can be prepared for viewing with a light microscope to show the different stages of meiosis in plant cells.

In the male reproductive organ of plants, meiosis takes place in cells known as pollen mother cells.

Fig. 4.1 and Fig. 4.2 show photomicrographs of two different stages of meiosis in pollen mother cells from a lily plant, *Lilium*.

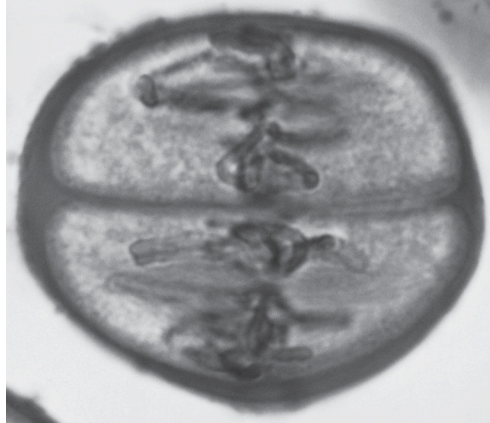


Fig. 4.1

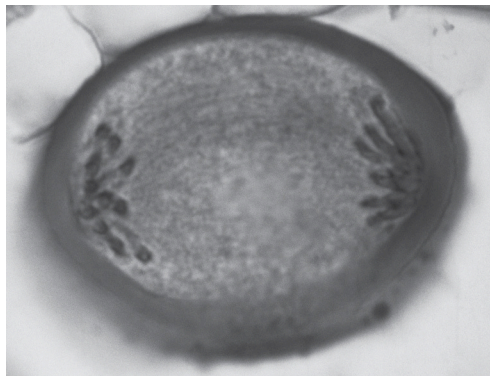


Fig. 4.2

- (i) Identify the stages of meiosis shown in Fig. 4.1 **and** Fig. 4.2.

Fig. 4.1

Fig. 4.2

[2]

- (ii) The cells formed at the end of meiosis in a *Lilium* pollen mother cell each have 12 chromosomes.

State the number of sister chromatids found in a *Lilium* pollen mother cell at the start of meiosis.

..... [1]

[Total: 8]